Chapter 4

RELATIONS

EXERCISE 4

Short Questions

Solve / write answers of the following short questions:

•Q.1(i) Define an equivalence relation. Write any equivalence relation defined on a set {1, 2, 3, 4}.

PU. 2012 (BS Math)

Sol: A relation on a set is called an equivalence relation if it is reflexive, symmetric, and transitive.

Two elements a and b that are related by an equivalence relation are called equivalent.

The notation $a \sim b$ is often used to denote that a and b are equivalent elements with respect to a particular equivalence relation.

Next consider the set {1, 2, 3, 4}.

The relation

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$$

is reflexive, symmetric and transitive, so it is equivalence relation.

(ii) Define Symmetric, Anti-symmetric and Transitive relations on a set.

PU. 2011 (BS Math)

Sol: **Symmetric Relation**

A relation R on a set A is called symmetric if $(b, a) \in R$ whenever $(a, b) \in R$, for all $a, b \in A$.

Anti-Symmetric Relation

A relation R on a set A such that for all $a, b \in A$, if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$ is called anti-symmetric.

Transitive Relation

A relation R on a set A is called transitive if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for all $a, b, c \in A$.

(iii) List the ordered pairs in the relation R from $A = \{0, 1, 2, 3, 4, 5\}$ to $B = \{0, 1, 2, 3\}$, where $(a, b) \in R$ if and only if $a = b$.

Sol: The required relation is

$$R = \{(0, 0), (1, 1), (2, 2), (3, 3)\}$$

(iv) **List the ordered pairs in the relation R from $A = \{0, 1, 2, 3, 4, 5\}$ to $B = \{0, 1, 2, 3\}$, where $(a, b) \in R$ if and only if $a \mid b$.**

Sol: The required relation is

$$R = \{(1, 0), (1, 1), (1, 2), (1, 3), (2, 0), (2, 2), (3, 0), (3, 3)\}$$

(v) **List the ordered pairs in the relation R from $A = \{0, 1, 2, 3, 4, 5\}$ to $B = \{0, 1, 2, 3\}$, where $(a, b) \in R$ if and only if $a + b = 5$.**

Sol: The required relation is

$$R = \{(0, 5), (1, 4), (2, 3), (3, 2)\}$$

(vi) **List the ordered pairs in the relation R from $A = \{0, 1, 2, 3, 4, 5\}$ to $B = \{0, 1, 2, 3\}$, where $(a, b) \in R$ if and only if $a > b$.**

Sol: The required relation is

$$R = \{(1, 0), (2, 0), (2, 1), (3, 0), (3, 1), (3, 2), (4, 0), (4, 1), (4, 2), (4, 3), (5, 1), (5, 2), (5, 3), (5, 4)\}$$

(vii) **List the ordered pairs in the relation R from $A = \{0, 1, 2, 3, 4, 5\}$ to $B = \{0, 1, 2, 3\}$, where $(a, b) \in R$ if and only if $a < b$.**

Sol: The required relation is

$$R = \{(0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (2, 3)\}$$

(viii) **List the ordered pairs in the relation R from $A = \{0, 1, 2, 3, 4, 5\}$ to $B = \{0, 1, 2, 3\}$, where $(a, b) \in R$ if and only if $\gcd(a, b) = 1$.**

Sol: The required relation is

$$R = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2), (4, 1), (4, 3), (5, 1), (5, 2), (5, 3)\}$$

(ix) **List all the ordered pairs in the relation $R = \{(a, b) \mid a \text{ divides } b\}$ on the set $\{1, 2, 3, 4, 5, 6\}$.**

Sol: The required relation is

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$$

(x) **Determine whether the relation R on the set of all people is reflexive, symmetric, anti-symmetric, and/or transitive, where $(a, b) \in R$ if and only if a is taller than b.**

Sol: Since $(a, a) \notin R$, so R is not reflexive.
If $(a, b) \in R$, then $(b, a) \notin R$, so R is not symmetric.
If $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$, so R is transitive.

(xi) **Determine whether the relation R on the set of all people is reflexive, symmetric, anti-symmetric, and/or transitive, where $(a, b) \in R$ if and only if a and b were born on the same day.**

Sol: It is reflexive, symmetric and transitive.

(xii) **Determine whether the relation R on the set of all real numbers is reflexive, where $(x, y) \in R$ if and only if $x = 1$ or $y = 1$.**

Sol: It is not reflexive.

(xiii) **Determine whether the relation R on the set of all real numbers is transitive, where $(x, y) \in R$ if and only if $x = 1$.**

Sol: It is not transitive.

(xiv) **Let $R_1 = \{(1, 2), (2, 3), (3, 4)\}$ and**
$R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (3, 4)\}$ be relations from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$. Find $R_1 \cap R_2$.

Sol: $R_1 \cap R_2 = \{(1, 2), (2, 3), (3, 4)\}$

(xv) **$R_1 = \{(a, b) \in R^2 \mid a > b\}$, the "greater than" relation,**
$R_2 = \{(a, b) \in R^2 \mid a < b\}$, the "less than" relation. Find $R_1 \cup R_2$.

Sol: Obviously,

$R_1 \cup R_2 = \{(a, b) \in R^2 \mid a > b \text{ or } a < b\}$

Long Questions

Q.2 **For each of these relations on the set $\{1, 2, 3, 4\}$, decide whether it is reflexive, whether it is symmetric, whether it is anti-symmetric, and whether it is transitive.**

(a) $\{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$

(b) $\{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$

Sol:(a) Not reflexive, not symmetric, it is transitive.

(b) Reflexive, symmetric.

Q.3 **Determine whether the relation R on the set of all real numbers is reflexive, symmetric, anti-symmetric, and/or transitive, where $(x, y) \in R$ if and only if**

(a) $x + y = 0$
(b) $x = \pm y$
(c) $x - y$ is a rational number
(d) $x = 2y$
(e) $xy \geq 0$
(f) $xy = 0$

Sol:(a) Symmetric, transitive.

(b) Reflexive, symmetric.

Q.4 **Let $R_1 = \{(1, 2), (2, 3), (3, 4)\}$ and**

$R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (3, 4)\}$ be relations from {1, 2, 3} to {1, 2, 3, 4}. Find

(a) $R_1 \cup R_2$ (b) $R_1 - R_2$ (d) $R_2 - R_1$

Sol:(a) $R_1 \cup R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (3, 4)\}$

(b) $R_1 - R_2 = \phi$

(c) $R_2 - R_1 = \{(1, 1), (2, 1), (2, 2), (3, 1), (3, 2), (3, 3)\}$

Q.5 List the 16 different relations on the set {0, 1}.

Sol: $R_1 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$

$R_2 = \{(0, 0), (0, 1), (1, 0)\}$ $R_3 = \{(0, 0), (0, 1), (1, 1)\}$

$R_4 = \{(0, 1), (1, 0), (1, 1)\}$ $R_5 = \{(0, 0), (1, 0), (1, 1)\}$

$R_6 = \{(0, 0), (0, 1)\}$ $R_7 = \{(0, 0), (1, 1)\}$

$R_8 = \{(1, 0), (1, 1)\}$ $R_9 = \{(1, 0), (0, 1)\}$

$R_{10} = \{(0, 0)\}$ $R_{11} = \{(1, 0)\}$

$R_{12} = \{(0, 1)\}$ $R_{13} = \{(1, 1)\}$ Etc.

Q.6 Which of these relations on {0, 1, 2, 3} are equivalence relations? Determine the properties of an equivalence relation that the others lack.

(i) $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$

(ii) $\{(0, 0), (0, 2), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$

(iii) $\{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$

(iv) $\{(0, 0), (1, 1), (1, 3), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$

(v) $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$

Sol:(i) Equivalence relation. (ii) Not reflexive.

(iii) Equivalence relation. (iv) Not transitive.

(v) Equivalence relation.

*****